

Do Aggregate Indices Mask Economic Disruptions?

Geopolitical Oil Shocks, Sectoral Heterogeneity, and Aggregation Masking: Evidence from a Multi-Event Framework

Oil Shock

4 Supply Shocks

Hormuz • Russia–Ukraine
OPEC+ • Middle East

1 Demand Shock

COVID-19

34 ETFs • 2020–2026

Sectoral Responses

Supply shocks:

▲ **High Oil Beta** → **Winners**

▼ **Low Oil Beta** → **Losers**

Demand shocks:

▼ **All sectors fall together**

Aggregate Index

S&P 500: **−1.81%**

Looks calm

≠

Reality

59 pp sectoral range

+42% to −17% across sectors

~95%

of sectoral
disruption
concealed

$\Psi = 0.947$

Aggregation
Masking Index

Masking is materially stronger during crises than non-event periods

Index-level responses understate economically meaningful heterogeneity for investors and policymakers